

Mithish Ravisankar

250-258-3091 | mithish.work@gmail.com | [linkedin.com/in/mithish17](https://www.linkedin.com/in/mithish17) | github.com/MithishR | mithishravisankar.com

EDUCATION

The University of British Columbia

Bachelor of Science in Computer Science, Minor in Data Science (Co-Op)

- Dean's List, Faculty of Science Int'l Student Award, Int'l Learning Programs Award, Harmandeep Kaur Int'l Student Award

Kelowna, British Columbia

Sep 2021 – May 2026

EXPERIENCE

Directed Studies - Research Software Developer

Sep 2025 – Present

University of British Columbia

Kelowna, British Columbia

- Engineered and scaled microservice-based cloud infrastructure for HelpMe, UBC's academic platform serving 10,000+ users; utilized Kubernetes and Docker to ensure high availability and zero service disruptions during peak traffic.
- Designed and implemented RESTful API architectures with a focus on fault tolerance (replication and redundancy), increasing system resilience and data availability during rigorous stress testing.
- Integrated GitHub Actions and automated quality gates to streamline the CI/CD pipeline, reducing deployment lead time and ensuring production-ready builds across Agile SCRUM sprints.
- Re-architected backend services using Python and MSSQL to improve modularity and decoupling, significantly reducing request latency and simplifying the integration testing of individual microservice components.
- Engineered automated ingestion workflows for microservices, reducing manual update effort by 70% and ensuring data synchronization across distributed backend components to maintain system-wide consistency.

Research Intern - Software Development

Jul 2025 – Aug 2025

Technical University of Munich

Munich, Germany

- Architected a custom Git-style versioning subsystem for the momenTUM Research Platform using Python, MongoDB (NoSQL) and RESTful API architecture, enabling end-to-end reproducible workflows and reducing verification time by 85% across multi-center trials.
- Enforced 100% audit alignment by applying semantic versioning and branching principles, mirroring industrial release management and security standards used in highly regulated sectors.
- Engineered custom diff-comparison logic and commit-history tracking using Node.js and Python, ensuring absolute data integrity and a transparent audit trail for large-scale clinical metadata..

Software Development Intern

May 2024 – Apr 2025

Equitable Insurance

Waterloo, Ontario

- Architected a full-stack stale pricing detection platform using Python, Flask, and SQL Server (MSSQL) within an Agile environment, reducing stale bond data by 95% and ensuring pricing accuracy across \$250 million in assets.
- Developed and integrated RESTful automation services via Azure Functions, Key Vault, and Teams API to secure credential delivery, accelerating internal workflows from hours to seconds.
- Engineered data-processing pipelines and Bloomberg integrations using Python (Pandas) and SQL, optimizing transformation logic to ensure high-availability reporting and error-free accounting analytics.
- Implemented CI/CD workflows and automated testing suites using pytest (unit/integration) and dependency management tools to maintain stable, production-ready deployments and code quality.
- Built and managed Power BI dashboards for cash flow forecasting and compliance, utilizing Flask and Go scripts to automate data validation and reduce manual verification effort by 60%.

PROJECTS

Fora - UBC Okanagan Task Marketplace | Python, Next.js, TypeScript, Supabase, FastAPI, Docker

- Architected a Full-Stack campus marketplace with Next.js and FastAPI, implementing RESTful APIs for real-time task posting, location-based discovery (Google Maps API), and secure Role-Based Access Control for 10,000+ potential users.
- Implemented transactional logic in PostgreSQL using database triggers to manage a credit-based rewards system, deploying the containerized backend to Azure using ARM templates to ensure consistent, reproducible environments.

TransitMate - Ride Sharing Platform | Java, Android Studio, Firebase, JUnit

- Developed a native Android application using Java, implementing modern UI/UX best practices and Human-Computer Interaction (HCI) principles to deliver a performant, customer-centric interface for 100+ potential users.
- Integrated Firebase backend services and Google Maps API to support real-time authentication and location discovery, utilizing Object-Oriented Programming (OOP) and JUnit to ensure modular, testable, and reliable mobile code.

TECHNICAL SKILLS

Languages: Java (Spring Boot), Python (FastAPI/Flask/Django), SQL (MSSQL/PostgreSQL), JavaScript/TypeScript, C++, R

Frameworks & Libraries: Spring Boot, FastAPI, Flask, Django, Express.js, Angular, React/Redux, Pandas, NumPy

Tools & Platforms: Azure DevOps, Power BI, Firebase, JUnit (Testing), Git, Android Studio

Cloud & APIs: Microsoft Azure, RESTful/SOAP APIs, Google Cloud Platform (GCP), Microsoft Teams API, GraphQL

Databases: SQL Server (MSSQL), PostgreSQL, MySQL, MongoDB, SQLite